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Profiling the Biomechanical Responses to Workload on the Human Myocyte to Explore the Concept of Myocardial Fatigue and Reversibility: Rationale and Design of the POWER Heart Failure Study

[Patrick Tran](#), [Adam Linekar](#), Uday Dandekar, Thomas Barker, Sendhil Balasubramanian, Jain Bhaskara-Pillai, [Sharn Shelley](#), [Helen Maddock](#), Prithwish Banerjee

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Abstract

It remains unclear why some patients develop heart failure without evidence of structural damage. One theory relates to impaired myocardial energetics and ventricular-arterial decoupling as the heart works against adverse mechanical load. In this original study, we propose the novel concept of myocardial fatigue to capture this phenomenon and aim to investigate this using human cardiomyocytes subjected to a modern work-loop contractility model that closely mimics in vivo cardiac cycles. This proof-of-concept study (NCT04899635) will use human myocardial tissue samples from patients undergoing cardiac surgery to develop a reproducible protocol to isolate robust calcium-tolerant cardiomyocytes. Thereafter, work-loop contractility experiments will be performed over a range of preload, afterload and cycle frequency as a function of time to elicit any reversible reduction in contractile performance (i.e. fatigue). This will provide novel insight into mechanisms behind heart failure and myocardial recovery and serve as a valuable research platform in translational cardiovascular research.

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Keywords

- Myocardial fatigue
- Cardiomyocytes
- Work-loops
- Contractility
- Afterload
- Preload

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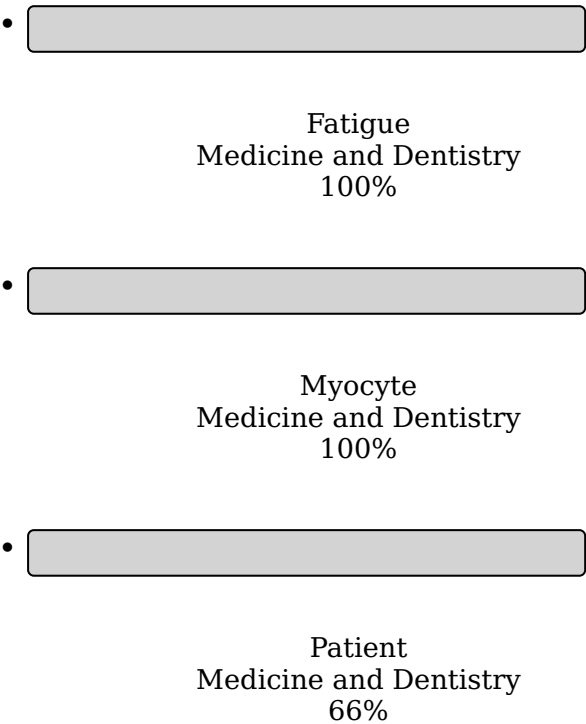
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